7/11/2025

OSS: VISTO CHE IL LIMITE È UN "CONCETTO LOCALE" POSSIAMO DEDURRE IL LIMITE DI FUNZIONI LOCALMENTE MONOTONE.

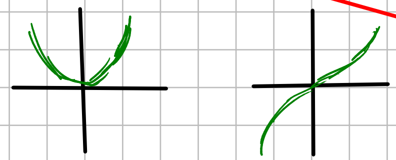
ES: $\lim_{x \to x_0} \cos(x) \to \cos(x_0) \quad \forall x_0 \in \mathbb{R}$ $\quad \lim_{x \to x_0} \sin(x) \to \sin(x_0) \quad \forall x_0 \in \mathbb{R}$ $\quad \lim_{x \to x_0} \tan(x) \to \tan(x_0) \quad \forall x_0 \neq \frac{\pi}{2} + k\pi$

$$\Downarrow$$

$$\lim_{x \to \pm\frac{\pi}{2}} \tan(x) \to \pm\infty$$

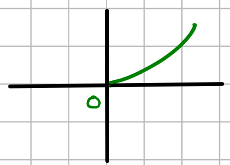
ALTRI ESEMPI: POTENZE : $n \in \mathbb{N},\ n \geq 1$

$$\lim_{x \to +\infty} x^n = +\infty \qquad \lim_{x \to -\infty} x^n = \begin{cases} +\infty & \text{SE } n \text{ È PARI} \\ -\infty & \text{SE } n \text{ È DISPARI} \end{cases}$$



$\alpha \in \mathbb{R}$

$$\lim_{x \to +\infty} x^\alpha = \begin{cases} +\infty & \alpha > 0 \\ 1 & \alpha = 0 \\ 0 & \alpha < 0 \end{cases} \qquad \lim_{x \to 0^+} x^\alpha = \begin{cases} 0 & \alpha > 0 \\ 1 & \alpha = 0 \\ +\infty & \alpha < 0 \end{cases}$$



$\alpha > 0$

$\alpha < 0$

LIMITI SENO E COSENO:

$$\nexists \lim_{x \to \pm\infty} \sin(x) \qquad \nexists \lim_{x \to \pm\infty} \cos(x)$$

FUNZIONI IPERBOLICHE:

SINH(x)

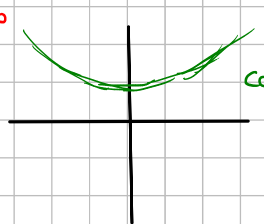
$$\lim_{x \to +\infty} \sinh(x) = \lim_{x \to +\infty} \frac{e^x - e^{-x}}{2} = \lim_{x \to +\infty} \frac{e^x - \overbrace{\frac{1}{e^x}}^{0}}{2} = \lim_{x \to +\infty} \frac{e^x}{2} = +\infty$$

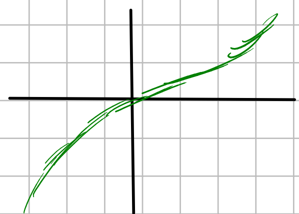
$$\lim_{x \to -\infty} \sinh(x) = \lim_{x \to -\infty} \frac{e^x - e^{-x}}{2} = \lim_{x \to -\infty} \frac{\overbrace{\frac{1}{e^x}}^{0} - e^x}{2} = \lim_{x \to -\infty} \frac{-e^x}{2} = -\infty$$

COSH(x)

$$\lim_{x \to +\infty} \cosh(x) = \lim_{x \to +\infty} \frac{e^x + e^{-x}}{2} = \lim_{x \to +\infty} \frac{e^x + \overbrace{\frac{1}{e^x}}^{0}}{2} = \lim_{x \to +\infty} \frac{e^x}{2} = +\infty$$

$$\lim_{x \to -\infty} \cosh(x) = \lim_{x \to -\infty} \frac{e^x + e^{-x}}{2} = \lim_{x \to -\infty} \frac{\overbrace{\frac{1}{e^x}}^{0} + e^x}{2} = \lim_{x \to -\infty} \frac{e^x}{2} = +\infty$$

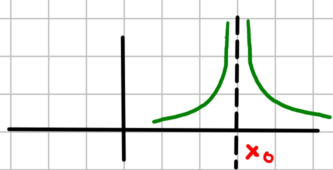




POTENZE CON ESPONENTE NEGATIVO:

$$f(x) = \frac{1}{(x-x_0)^n} \qquad n \in \mathbb{N},\ x_0 \in \mathbb{R}$$

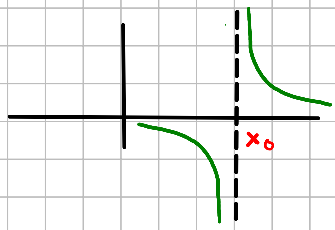
$$\lim_{x \to x_0^+} \frac{1}{(x-x_0)^n} = +\infty \qquad \lim_{x \to x_0^+} \frac{1}{(x-x_0)^n} = +\infty$$



n PARI

$$\lim_{x \to x_0^+} \frac{1}{(x-x_0)^m} = +\infty \qquad \lim_{x \to x_0^+} \frac{1}{(x-x_0)^m} = -\infty$$

$$\nexists \lim_{x \to x_0} \frac{1}{(x-x_0)^m}$$



m DISPARI

ES: • $\lim_{x \to +\infty} \dfrac{4x^3 - 2x^2 + 6x}{6x^2 - 10x}$ " $\dfrac{\infty}{\infty}$ "

$$\lim_{x \to +\infty} \frac{4x^3 - 2x^2 + 6x}{6x^2 - 10x} = \lim_{x \to +\infty} \frac{\cancel{x^3}\left(4 - \overbrace{\frac{2}{x}}^{0} + \overbrace{\frac{6}{x^2}}^{0}\right)}{\cancel{x^2}\left(6 - \underbrace{\frac{10}{x}}_{0}\right)} = \frac{2}{3}x = +\infty$$

• $$\lim_{x \to -\infty} \frac{-x^4 + 7x^2}{2x^4 + 1} = \lim_{x \to -\infty} \frac{\cancel{x^4}\overbrace{\left(-1 + \frac{7}{x^2}\right)}^{-1}}{\cancel{x^4}\underbrace{\left(2 + \frac{1}{x^4}\right)}_{2}} = -\frac{1}{2}$$

$$\frac{1}{2} + x = \frac{1}{5} \Rightarrow x = \frac{1}{5} - \frac{1}{2} = -\frac{3}{10}$$

• $$\lim_{x \to +\infty} \frac{\sqrt[5]{x} + \sqrt[4]{x} - 2\sqrt{x}}{x^2 - \sqrt[6]{x}} = \lim_{x \to +\infty} \frac{x^{\frac{1}{5}} + x^{\frac{1}{4}} - 2x^{\frac{1}{2}}}{x^{\frac{1}{2}} - x^{\frac{1}{6}}} = \lim_{x \to +\infty} \frac{x^{\frac{1}{2}}\left(\overbrace{x^{-\frac{3}{10}}}^{0} + \overbrace{x^{-\frac{1}{4}}}^{0} - 2\right)}{x^2\left(1 - \underbrace{x^{-\frac{11}{6}}}_{0}\right)} =$$

$$\lim_{x \to +\infty} x^{-\frac{3}{2}}(-2) = \lim_{x \to +\infty} \frac{-2}{x^{\frac{3}{2}}} = 0$$

RACCOLGO LA x CON ESPONENTE + ALTO.

ES: • $\lim_{x \to 0} \dfrac{-3x^3 + 4x^2 + 6x}{5x^4 - 2x^2 + 3x}$ " $\dfrac{0}{0}$ "

$$\lim_{x \to 0} \frac{-3x^3 + 4x^2 + 6x}{5x^4 - 2x^2 + 3x} = \lim_{x \to 0} \frac{\cancel{x}(\overbrace{-3x^2 + 4x}^{0} + 6)}{\cancel{x}(\underbrace{5x^3 - 2x}_{0} + 3)} = \frac{6}{3} = 2$$

• $$\lim_{x \to 0^+} \frac{x^2 - x^3 + \sqrt{x}}{x^2 - x^{3/2} + \sqrt[4]{x}} = \lim_{x \to 0^+} \frac{\overset{x^{\frac{1}{4}}}{\cancel{x^{\frac{1}{2}}}}(\overbrace{x^{3/2} + x^{5/2}}^{0} + 1)}{\cancel{x^{1/4}}(\underbrace{x^{7/4} - x^{5/4}}_{0} + 1)} = \lim_{x \to 0^+} x^{\frac{1}{4}} = 0$$

RACCOLGO LA x CON ESPONENTE + PICCOLO.

ES: • $\lim_{x \to +\infty} \sqrt{x^2 + 1} - \sqrt{x^2 - 1}$ " $\infty - \infty$ "

$$\lim_{x \to +\infty} \sqrt{x^2+1} - \sqrt{x^2-1} = \lim_{x \to +\infty} |x|\sqrt{1 + \frac{1}{x^2}} - |x|\sqrt{1 - \frac{1}{x^2}} = \lim_{x \to +\infty} |x| \underbrace{\left(\overbrace{\sqrt{1 + \frac{1}{x^2}}}^{1} - \overbrace{\sqrt{1 - \frac{1}{x^2}}}^{1}\right)}_{0}$$

$$\lim_{x \to +\infty} x \underbrace{\left(\overbrace{\sqrt{1 + \frac{1}{x^2}}}^{1} - \overbrace{\sqrt{1 - \frac{1}{x^2}}}^{1}\right)}_{0} \Rightarrow \text{" } \infty \cdot 0 \text{ "}$$

NO!

• $$\lim_{x \to +\infty} \sqrt{x^2+1} - \sqrt{x^2-1} \cdot \frac{\sqrt{x^2+1} + \sqrt{x^2-1}}{\sqrt{x^2+1} + \sqrt{x^2-1}} = \frac{\overbrace{\cancel{x^2} + 1 - (\cancel{x^2} - 1)}^{2}}{\underbrace{\sqrt{x^2+1} + \sqrt{x^2-1}}_{+\infty}} = 0$$

• $$\lim_{x \to +\infty} \sqrt{x^2+1} - \sqrt{3x^2-1} \quad \text{" } \infty - \infty \text{ "} = \lim_{x \to +\infty} \underbrace{|x|}_{+\infty} \underbrace{\left(\sqrt{1 + \frac{1}{x^2}} - \sqrt{3 - \frac{1}{x^2}}\right)}_{1 - \sqrt{3} < 0} = -\infty$$

LIMITI DELLE FUNZIONI COMPOSTE: $A \subseteq \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ DI ACC. PER A, $f: A \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$.

SUPPONIAMO: 1) $\lim\limits_{x \to x_0} f(x) = y_0 \in \mathbb{R}$ 2) $\lim\limits_{y \to y_0} g(y) = \ell \in \mathbb{R}$ 3) $f(x) = y_0$ DEF. PER $x \to x_0$

ALLORA: $\boxed{\lim\limits_{x \to x_0} g(f(x)) = \ell}$

ES: • $\lim\limits_{x \to +\infty} \sin(e^{-x}) = \lim\limits_{y \to 0} \sin(y) = 0$

$\boxed{\begin{array}{l} e^{-x} = y \to 0 \quad x \to +\infty \\ g(y) = \sin(y) \end{array}}$

• $\lim\limits_{x \to 0} \dfrac{\sin(2x)}{x} = \lim\limits_{x \to 0} \dfrac{\sin(2x)}{2x} \cdot 2 \Rightarrow 2x = y \Rightarrow \lim\limits_{y \to 0} \underbrace{\dfrac{\sin(y)}{y}}_{1} \cdot 2 = 2$

• $\lim\limits_{x \to 0} \dfrac{3\sin^3(x) + \sin(x)}{\sin^4(x) + \sin^3(x)} \Rightarrow \sin(x) = y \Rightarrow \lim\limits_{y \to 0} \dfrac{3y^3 + y}{y^4 + y^3} = \lim\limits_{y \to 0} \dfrac{\cancel{y}\overbrace{(3y^2+1)}^{1}}{y^{\cancel{3}\,2}\underbrace{(y+1)}_{1}} = \dfrac{1}{y^2} = +\infty$

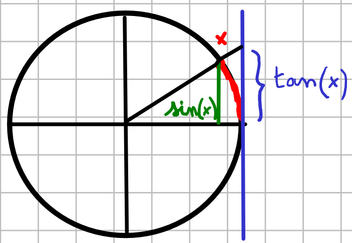
• $\lim\limits_{x \to 0^+} \dfrac{\log(x) + 1}{\log(x) + 2} \Rightarrow \log(x) = y \Rightarrow \lim\limits_{y \to -\infty} \dfrac{y+1}{y+2} = 1$

LIMITI FONDAMENTALI:

1)

$\boxed{\lim\limits_{x \to 0} \dfrac{\sin(x)}{x} = 1}$

DIM: PER $x \in (0, \frac{\pi}{2})$ ABBIAMO: $\boxed{\sin(x) < x < \tan(x)}$



$\Downarrow$ DIVIDIAMO PER $\sin(x)$

$1 < \dfrac{x}{\sin(x)} < \dfrac{1}{\cos(x)}$ OVVERO:

$\underbrace{\cos(x)}_{\substack{x \to 0^+ \\ 1}} < \dfrac{\sin(x)}{x} < \underbrace{1}_{\substack{x \to 0^+ \\ 1}}$

PER IL TEOREMA DEI 2 CARABINIERI: $\boxed{\lim\limits_{x \to 0^+} \dfrac{\sin(x)}{x} = 1}$ E DATO CHE È PARI $\Rightarrow$ $\boxed{\lim\limits_{x \to 0^-} \dfrac{\sin(x)}{x} = 1}$

2) $\boxed{\lim\limits_{x \to 0} \dfrac{1 - \cos(x)}{x^2} = \dfrac{1}{2}} \Rightarrow \lim\limits_{x \to 0} \dfrac{1 - \cos(x)}{x^2} \cdot \dfrac{1 + \cos(x)}{1 + \cos(x)} = \lim\limits_{x \to 0} \dfrac{1 - \cos^2(x)}{x^2} \cdot \dfrac{1}{1 + \cos(x)} =$

$\boxed{\cos^2 x + \sin^2(x) = 1} \Rightarrow \lim\limits_{x \to 0} \underbrace{\dfrac{\sin^2(x)}{x^2}}_{1^2} \cdot \underbrace{\dfrac{1}{1 + \cos(x)}}_{\frac{1}{2}} = \dfrac{1}{2}$

3) $\boxed{\lim\limits_{x \to 0} \dfrac{\tan(x)}{x} = 1} \Rightarrow \lim\limits_{x \to 0} \dfrac{\tan(x)}{x} = \lim\limits_{x \to 0} \dfrac{\sin(x)}{x} \cdot \dfrac{1}{\cos(x)} = 1$

4) $\boxed{\lim\limits_{x \to 0} \dfrac{\arctan(x)}{x} = 1} \Rightarrow \lim\limits_{x \to 0} \dfrac{\arctan(x)}{x} \Rightarrow \begin{array}{l} y = \arctan(x) \\ x = \tan(y) \end{array} \Rightarrow \lim\limits_{y \to 0} \dfrac{y}{\tan(y)} = 1$

5) $\boxed{\lim_{x\to 0} \frac{\arcsin(x)}{x} = 1}$ $\Rightarrow \lim_{x\to 0} \frac{\arcsin(x)}{x} \Rightarrow$ $y = \arcsin(x)$, $x = \sin(y)$ $\Rightarrow \lim_{y\to 0} \frac{y}{\sin(y)} = 1$

6) $\boxed{\lim_{x\to +\infty} \left(1+\frac{1}{x}\right)^x = e}$ $=$ $\boxed{\lim_{x\to -\infty} \left(1+\frac{1}{x}\right)^x = e}$

7) $\boxed{\lim_{x\to +\infty} \left(1+\frac{a}{x}\right)^x = e^a}$ $\Rightarrow \lim_{x\to +\infty} \left(1+\frac{a}{x}\right)^x = \lim_{x\to +\infty} \left(\left(1+\frac{a}{x}\right)^{\frac{x}{a}}\right)^a = e^a$ $\quad a \in \mathbb{R}$

8) $\boxed{\lim_{x\to 0^+} (1+ax)^{\frac{1}{x}} = e^a}$ $\Rightarrow \lim_{x\to 0^+} (1+ax)^{\frac{1}{x}} \Rightarrow$ $y = \frac{1}{x}$, $x = \frac{1}{y}$ $\Rightarrow \lim_{y\to +\infty} \left(1+\frac{a}{y}\right)^y = e^a$ $\quad a \in \mathbb{R}$

9) $\boxed{\lim_{x\to 0^+} \frac{\log(1+x)}{x} = 1}$ $\Rightarrow \lim_{x\to 0^+} \frac{\log(1+x)}{x} = \lim_{x\to 0^+} \frac{1}{x}\log(1+x) = \lim_{x\to 0^+} \log\left((1+x)^{\frac{1}{x}}\right) =$

$= \log(e) = 1$ LOGARITMO

10) $\boxed{\lim_{x\to 0} \frac{e^x - 1}{x} = 1}$ $\Rightarrow \lim_{x\to 0} \frac{e^x-1}{x} \Rightarrow$ $y = e^x - 1$, $x = \log(y+1)$ $\Rightarrow \lim_{y\to 0} \frac{y}{\log(y+1)} = 1$ ESPONENZIALE

11) $a>0,\ a\neq 1 \Rightarrow$ $\boxed{\lim_{x\to 0} \frac{\log_a(1+x)}{x} = \log_a(e)}$ $\Rightarrow \lim_{x\to 0} \frac{\log_a(1+x)}{x} = \lim_{x\to 0} \overbrace{\frac{\log(1+x)}{x}}^{1} \cdot \log_a(e) =$

$= \log_a(e)$

12) $a>0,\ a\neq 1 \Rightarrow$ $\boxed{\lim_{x\to 0} \frac{a^x - 1}{x} = \log(a)}$ $\Rightarrow \lim_{x\to 0} \frac{a^x-1}{x} = \lim_{x\to 0} \frac{e^{\log(a^x)} - 1}{x} =$

$= \lim_{x\to 0} \underbrace{\frac{e^{x\cdot\log(a)} - 1}{x\cdot\log(a)}}_{1} \cdot \log(a) = \log(a)$

13) $\alpha \in \mathbb{R},\ \alpha \neq 0$ $\boxed{\lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha}$ $\Rightarrow \lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \lim_{x\to 0} \frac{e^{\alpha\log(1+x)} - 1}{x}$ $\Rightarrow y = \log(1+x)$, $x = e^y - 1$

$\Rightarrow \lim_{y\to 0} \frac{e^{\alpha y} - 1}{e^y - 1} \cdot \frac{\alpha y}{\alpha y} = \lim_{y\to 0} \underbrace{\frac{e^{\alpha y} - 1}{\alpha y}}_{1} \cdot \frac{\alpha y}{e^y - 1} \to 1 = \alpha$